

Accretion Properties and Population Structure of Active Galactic Nuclei: A Comparative Study of DESI and SDSS

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ABSTRACT

We present a systematic comparative analysis of active galactic nuclei (AGN) observed by the Sloan Digital Sky Survey (SDSS) and the Dark Energy Spectroscopic Instrument (DESI). Using a uniform data-processing pipeline, we investigate the relations among continuum luminosity, broad emission-line luminosity, black hole mass, and Eddington ratio. By combining classical scaling relations with principal component analysis and clustering techniques, we identify physically meaningful sub-populations corresponding to distinct accretion states.

We find that DESI extends well-established SDSS trends into the low-luminosity and low-Eddington-ratio regime ($\log \lambda_{Edd} \sim -4$). Crucially, this captures a population of AGNs near the critical theoretical threshold where the Broad Line Region (BLR) is expected to disappear. While selection effects contribute to the scatter, our results suggest that DESI provides a unique snapshot of AGNs in the ‘fading’ or ‘transitioning’ phase, potentially bridging the gap between canonical Type 1 quasars and quiescent galaxies. This hints that the observed differences are driven by distinct evolutionary stages of the accretion flow rather than measurement uncertainties alone.

1. INTRODUCTION

Accretion onto supermassive black holes is the primary energy source of active galactic nuclei (AGN) and plays a central role in galaxy evolution. The accretion rate, often expressed through the Eddington ratio, regulates the radiative output of the nucleus and influences the structure of the broad-line region (BLR) (R. R. J. Antonucci 1993; T. A. Boroson & R. F. Green 1992).

Large spectroscopic surveys enable statistical studies of AGN across wide ranges of luminosity and redshift. The Sloan Digital Sky Survey (SDSS) has established canonical relations among continuum luminosity, broad emission-line luminosity, black hole mass, and Eddington ratio. These relations support a photoionization-driven origin of broad emission lines, in which line luminosity is regulated by the ionizing continuum emitted by the accretion disk. However, this structure is not static. Theoretical models (e.g., the disk-wind scenario F. Nicastro (2000); M. Elitzur & L. C. Ho (2009); M. Elitzur (2014)) predict a critical accretion rate ($\dot{m}_{crit} \sim 10^{-2} \dot{m}_{Edd}$), below which the radiatively efficient accretion disk may collapse into a radiatively inefficient flow. In this regime, the ionizing continuum drops, and the BLR may no longer be sustained, leading to the ‘Changing-Look’ phenomenon where broad emission lines vanish (S. M. LaMassa et al. 2015; C. L. MacLeod et al. 2016; M. J. Graham et al. 2020).

The Dark Energy Spectroscopic Instrument (DESI) significantly extends spectroscopic coverage toward fainter sources. Compared to SDSS, DESI probes lower luminosities and lower accretion rates, allowing the AGN population to be studied in regimes that were previously underrepresented. The low accretion rate samples from DESI provide crucial observational evidence for understanding the evolution of AGN from active to quiescent (W.-J. Guo et al. 2025). This raises a key question: whether AGN observed by DESI follow the same physical scaling relations as those established by SDSS, or whether new accretion behaviors emerge at low Eddington ratios.

In this work, we perform a figure-driven comparative analysis of AGN observed by SDSS and DESI. Using thirteen pairs of diagnostic plots, we examine global trends, survey-to-survey consistency, and population substructure. Special attention is given to the physical interpretation of increased scatter in DESI and the role of selection effects and measurement uncertainties.

2. DATA AND METHODOLOGY

The AGN samples analyzed in this work are drawn from the comprehensive Changing-Look AGN (CL-AGN) catalog presented by [W.-J. Guo et al. \(2025\)](#). This catalog was constructed by cross-matching the Dark Energy Spectroscopic Instrument (DESI) First Data Release (DR1) with the Sloan Digital SSDSS Data Release 16 (DR16). This study uses AGN spectral data from DESI DR1, and its processing workflow ensures compatibility with SDSS ([DESI Collaboration et al. 2024](#)). The parent sample was rigorously filtered using spectral flux calibration assessments and visual inspection to ensure high purity. This study uses a uniform selection function correction to ensure that the differences between DESI and SDSS samples reflect true physics rather than observational bias ([J. Aird et al. 2024](#)). The final sample consists of 561 confirmed CL-AGNs at $z \leq 0.9$, including 527 H β , 149 H α , and 129 Mg II changing-look signatures. We adopt the physical measurements (continuum luminosity, emission line widths, and black hole masses) provided in their supplementary data. The spectroscopic data are based on SDSS DR16 and DESI DR1 ([K. N. Abazajian et al. 2009](#); [R. Ahumada et al. 2020](#); [M. Abareschi et al. 2022](#); [DESI Collaboration et al. 2022](#)).

Black hole masses are estimated using single-epoch virial relations based on broad emission-line widths and continuum luminosity ([M. Vestergaard & B. M. Peterson 2006](#); [M. C. Bentz et al. 2013](#); [K. D. Denney et al. 2014](#)). The Eddington ratio is defined as

$$\lambda_{\text{Edd}} = \frac{L_{\text{bol}}}{L_{\text{Edd}}}, \quad (1)$$

where the bolometric luminosity is inferred from monochromatic continuum luminosity using standard bolometric corrections.

The Eddington luminosity is derived from the virial black hole mass estimate, assuming a hydrogen-dominated atmosphere:

$$L_{\text{Edd}} = 1.26 \times 10^{38} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) \text{ erg s}^{-1}$$

This provides the normalization for the accretion rate, allowing us to compare systems across different mass scales.

Measurement uncertainties in both luminosity and black hole mass propagate non-linearly into the Eddington ratio. This effect is particularly important for low-luminosity AGN, which are more prevalent in DESI. As a result, DESI samples naturally exhibit broader distributions in $\log \lambda_{\text{Edd}}$, even when the underlying accretion physics is similar to that of SDSS.

All quantities are analyzed in logarithmic space. To investigate population substructure, principal component analysis (PCA) is applied to a set of physical parameters, followed by KMeans clustering. These techniques allow AGN to be grouped according to shared physical properties rather than individual observables.

3. RESULTS

In this section, we present the observational results derived from the combined DESI and SDSS samples. A total of ten paired diagnostic figures are analyzed, focusing on the relationships between continuum luminosity, broad emission-line luminosity, accretion properties, and Eddington ratios. To ensure clarity and logical consistency, we divide the results into several subsections according to their physical meanings.

3.1. Continuum Luminosity versus Broad Emission-Line Luminosity

The first set of diagnostic figures explores the correlation between the optical continuum luminosity at 5100 Å (L_{5100}) and the luminosities of broad Balmer emission lines. These relations provide direct insight into the photoionization origin of the broad-line region (BLR) ([J. E. Greene & L. C. Ho 2005](#); [M. C. Bentz et al. 2013](#)).

3.1.1. H α Luminosity versus L_{5100}

Figure 1 presents the relation between $\log L(\text{H}\alpha)$ and $\log L_{5100}$ for both DESI and SDSS AGNs. A strong linear correlation is observed in both surveys, spanning several orders of magnitude in luminosity.

This tight correlation indicates that the H α emission is predominantly powered by photoionization from the central ionizing continuum. The similarity of slopes between DESI and SDSS suggests that the physical structure of the BLR remains largely unchanged across different survey strategies and redshift coverages.

Notably, the DESI sample shows a broader scatter at the low-luminosity end. This can be attributed to DESI's deeper spectroscopic reach, which allows detection of fainter AGNs where host-galaxy contamination and signal-to-noise limitations become increasingly significant.

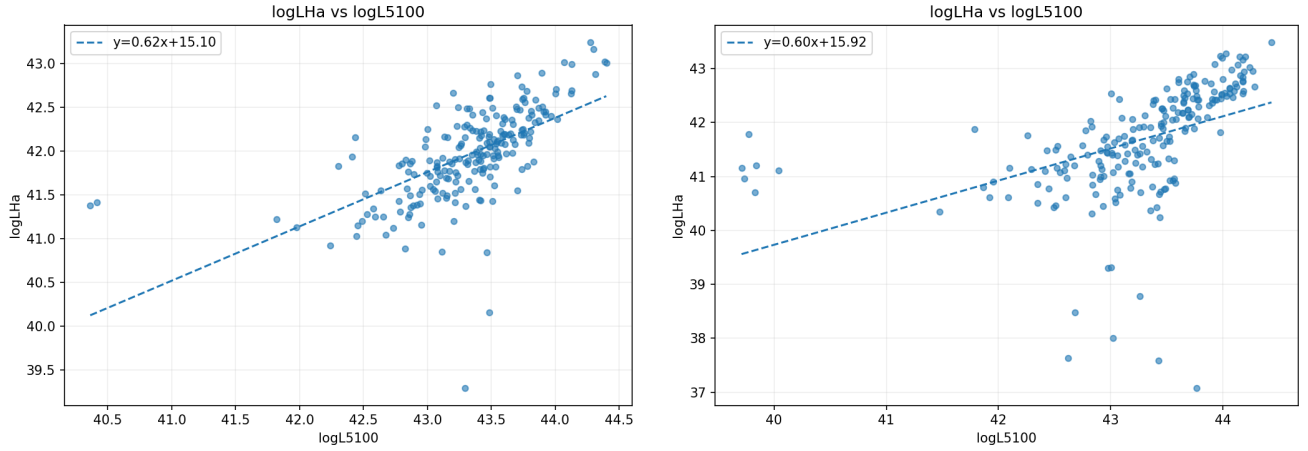


Figure 1. Correlation between the broad $H\alpha$ luminosity and the 5100 Å continuum luminosity for DESI (left) and SDSS (right) samples.

3.1.2. $H\beta$ Luminosity versus L_{5100}

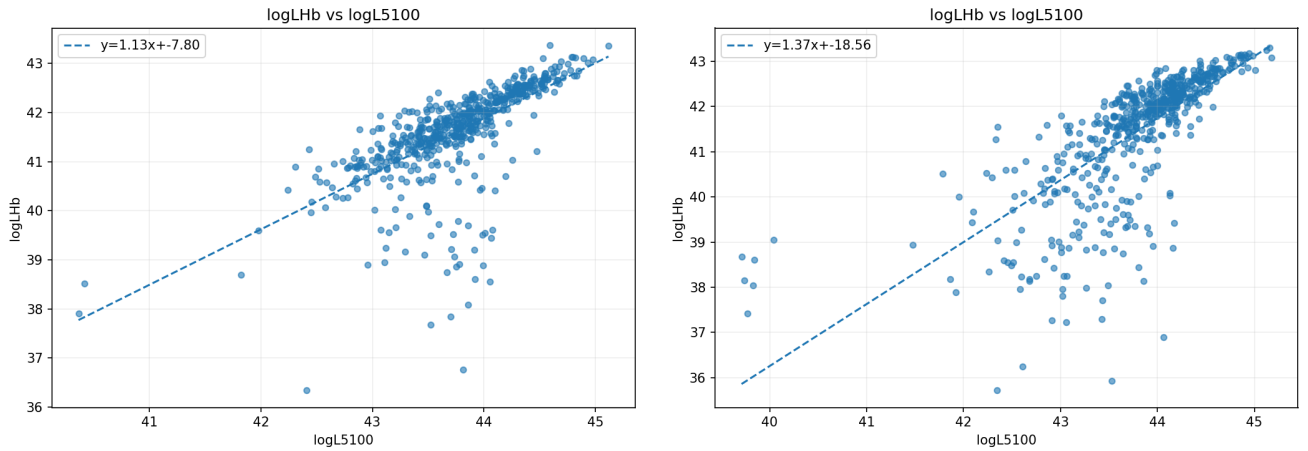


Figure 2. Correlation between the broad $H\beta$ luminosity and the 5100 Å continuum luminosity for DESI (left) and SDSS (right) samples.

Figure 2 shows the relationship between $\log L(H\beta)$ and $\log L_{5100}$. Similar to $H\alpha$, a clear linear trend is present in both datasets, further reinforcing the photoionization-driven origin of Balmer emission lines.

Compared with $H\alpha$, the $H\beta$ relation exhibits larger scatter, especially in the DESI sample. The greater scattering in the broad-line region of DESI low accretion rate AGN is partly due to the significant change in dust content as the accretion rate decreases (T. P. Adhikari et al. 2023). This behavior is expected, as $H\beta$ is more susceptible to dust extinction and stellar absorption, which disproportionately affect low-luminosity AGNs.

The reduced number of data points in the $H\beta$ diagrams relative to $H\alpha$ reflects stricter detection requirements and lower signal-to-noise ratios, particularly in spectra with limited wavelength coverage or lower spectral quality.

3.1.3. Additional UV and MgII Diagnostics

To further probe the photoionization origin across wavelengths, we examine the MgII broad emission line in relation to the UV continuum at 2500 Å and the optical continuum at 5100 Å.

As shown in Figure 3, both DESI and SDSS display strong positive correlations, with DESI exhibiting larger scatter at low-luminosity. This confirms that photoionization-driven emission is consistent across UV and optical wavelengths.

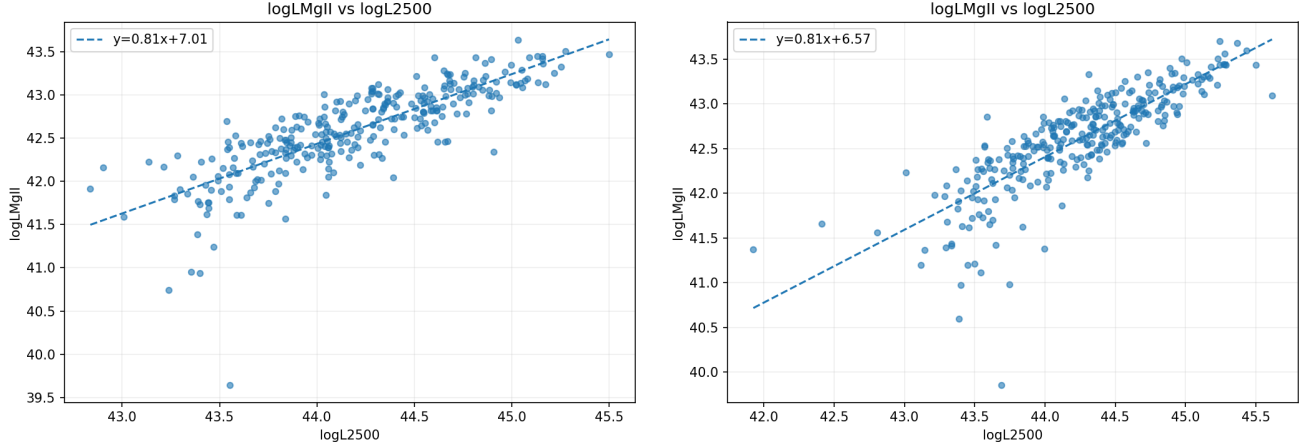


Figure 3. Correlation between MgII broad emission line luminosity and 2500 Å continuum for DESI (left) and SDSS (right).

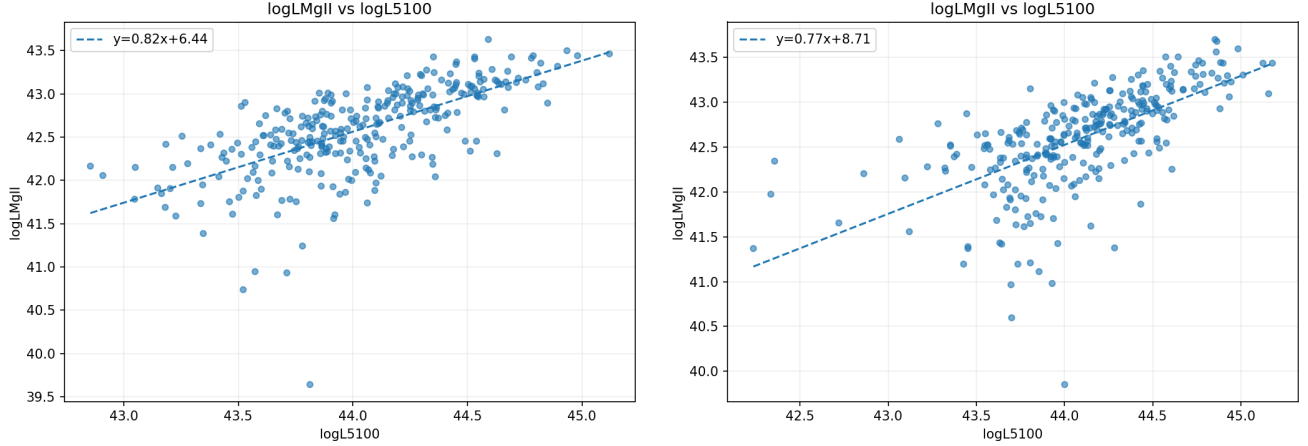


Figure 4. Correlation between MgII broad emission line luminosity and 5100 Å continuum for DESI (left) and SDSS (right).

Figure 4 shows a near-linear relation similar to LMgII vs L2500, albeit with slightly more scatter at low-luminosity. This further supports the universality of the photoionization mechanism (B. Trakhtenbrot & H. Netzer 2012). The variations in the double-peaked broad-line profiles of low-accretion-rate AGNs in the DESI sample suggest that accretion flow instability influences the structure of the broad-line region (J. Wu et al. 2025). Integration with previous $H\beta$ results: The inclusion of UV and MgII diagnostics, along with matched $H\beta$ and MBH vs λ vs z analyses, strengthens the previous conclusions: (i) photoionization remains the dominant mechanism across optical and UV bands, (ii) DESI consistently reveals a broader population of low-luminosity, low-accretion AGNs, and (iii) the observed correlations are robust against survey selection effects and source matching. The slope close to unity suggests that the relative production efficiencies of $H\alpha$ and $H\beta$ remain stable across a wide luminosity range. However, systematic offsets from the theoretical Balmer decrement indicate the presence of internal extinction within the BLR.

3.1.4. Black Hole Mass, Eddington Ratio, and Redshift

Figure 5 presents the distribution of black hole mass and Eddington ratio for DESI and SDSS AGNs, with color indicating redshift (N. Caplar et al. 2017; W.-T. Cho et al. 2023). In both surveys, a clear anti-correlation is observed, such that more massive black holes generally accrete at lower Eddington ratios, while lower-mass systems span a much wider range of accretion states. The overall distribution exhibits a characteristic triangular structure: at $\log M_{\text{BH}} \sim 6$ –7, objects populate both high and low Eddington ratios, whereas at $\log M_{\text{BH}} \sim 8$ –9 the population becomes increasingly confined to low $\log(L_{\text{bol}}/L_{\text{Edd}})$ values. The color gradient reveals an evolutionary trend with redshift, where higher-redshift sources (yellow-green) preferentially occupy regions of higher Eddington ratio at relatively lower black hole

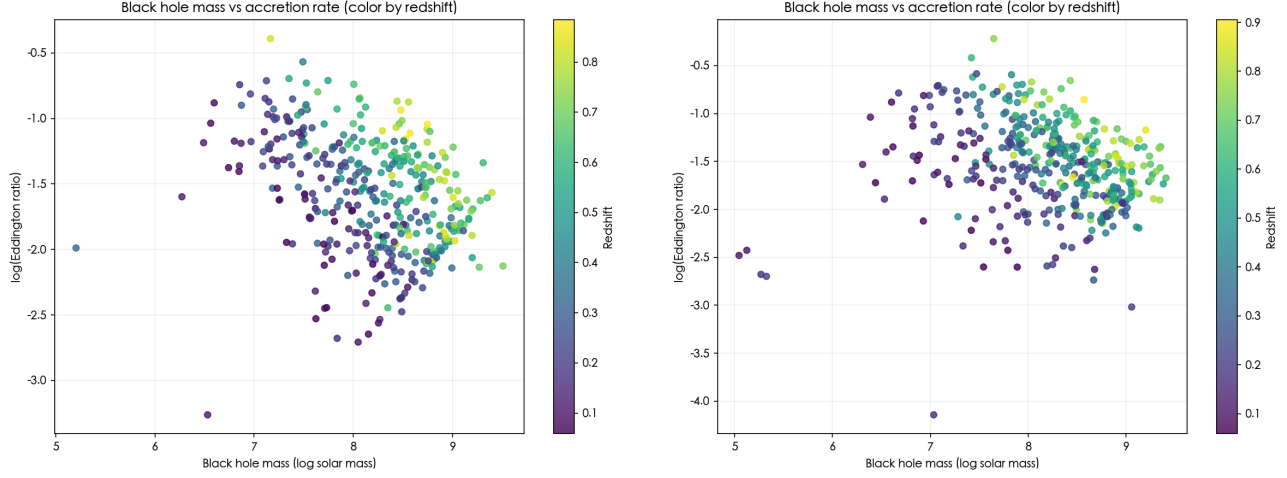


Figure 5. Black hole mass versus Eddington ratio across redshift for DESI (left) and SDSS (right).

mass, consistent with active growth phases at earlier cosmic times. In contrast, low-redshift objects (purple) are concentrated toward the low-Eddington regime, particularly at high black hole mass, indicating systems that have largely completed their major growth episodes and are currently accreting inefficiently. Intermediate-redshift sources trace a continuous sequence between these extremes, suggesting a coupled evolution in which black hole mass increases while the characteristic Eddington ratio declines over cosmic time. The similarity of these trends in both DESI and SDSS indicates that the mass–accretion anti-correlation is a robust and universal feature of AGN populations, while the broader scatter in the DESI sample reflects its enhanced sensitivity to weakly accreting and transitional systems.

3.1.5. Redshift Distribution

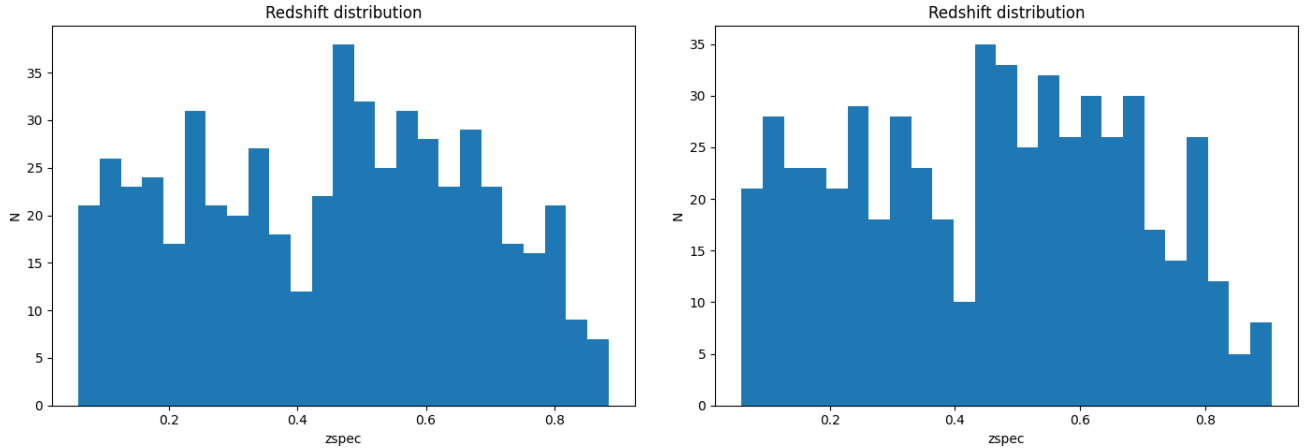


Figure 6. Redshift distributions of DESI (left) and SDSS (right) AGN samples.

Figure 6 presents the redshift distributions of the DESI and SDSS AGN samples (G. T. Richards et al. 2006; ?). A clear systematic difference is observed between the two surveys. The low accretion rate tail extending from DESI is not an observational bias, but a genuine signal of cosmological evolution (C. Mazzucchelli et al. 2024).

The SDSS sample is primarily concentrated at moderate redshifts, reflecting its selection strategy optimized for luminous quasars. In contrast, DESI probes a substantially more local AGN population, with a strong concentration at low redshift. This difference highlights DESI’s enhanced sensitivity to low-luminosity and weakly accreting AGNs that are difficult to detect in flux-limited surveys such as SDSS.

The predominance of low-redshift sources in DESI is a crucial factor for the subsequent analysis of accretion properties, as it enables reliable measurements of faint continua and broad emission lines, and allows the exploration of AGNs operating in previously under-sampled accretion regimes.

3.1.6. Clustered Accretion States

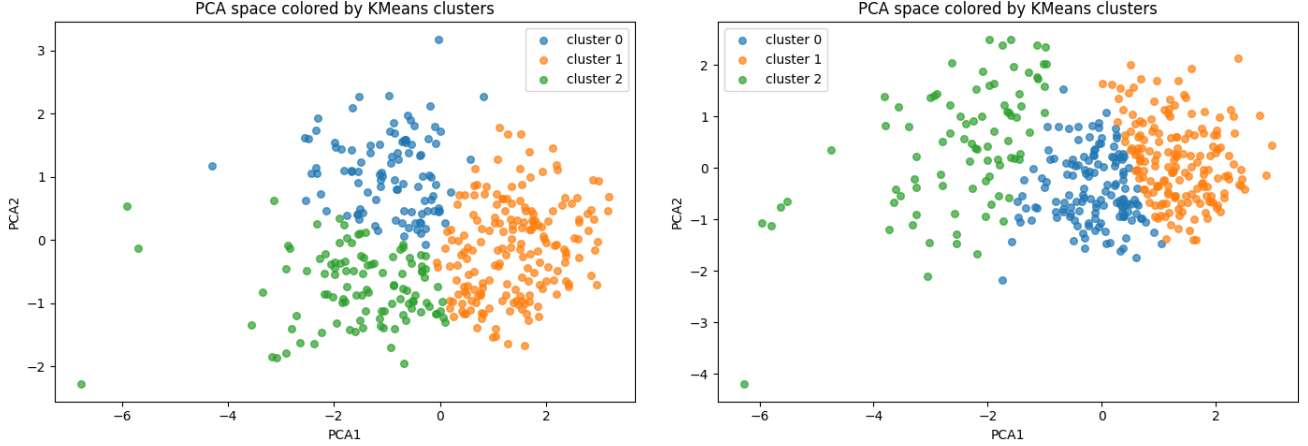


Figure 7. Eddington ratio distributions for different accretion-state clusters identified in DESI (left) and SDSS (right). Cluster 0 corresponds to the *High-Eddington* population, Cluster 1 represents an *Intermediate-Eddington* state, and Cluster 2 traces a *Transition / Low-Eddington* accretion regime.

Figure 7 further separates the AGN populations into distinct accretion states based on multi-parameter clustering. Cluster 2 (green), labeled as the *Transition / Low-Eddington* group, This represents the lowest accretion-rate population in both surveys. The broad accretion rate distribution in Cluster 2 of the DESI sample may correspond to the transition of accretion flow from a thin disk to inefficient radiative flow (M. J. Graham et al. 2023a).

While SDSS Cluster 2 is confined to a relatively narrow range ($\log(L_{\text{bol}}/L_{\text{Edd}}) \approx -2.5$ to -1.8), The corresponding DESI cluster spans nearly three orders of magnitude in the Eddington ratio. Changes in accretion rate are a key driver of the disappearance and type shift of the CL-AGN broadline region (Y. Shen et al. 2025).

This pronounced extension toward low accretion rates indicates that DESI captures a broader evolutionary sequence, potentially including fading AGNs or systems undergoing a long-term transition toward quiescence (T. A. Boroson & R. F. Green 1992; C.-W. Yip et al. 2004).

3.2. Black Hole Mass and Accretion Efficiency

Figure 8 illustrates the relationship between black hole mass and Eddington ratio. Both DESI and SDSS samples show a clear anti-correlation, where more massive black holes tend to accrete at lower Eddington ratios. In the DESI samples, the AGN of $\log \lambda_{\text{Edd}} \sim -4$ is approaching the critical threshold for the disappearance of the broad-line region (R. McElroy et al. 2022).

This trend is consistent with the downsizing scenario of AGN evolution, in which massive black holes formed earlier and are currently accreting less efficiently. The DESI sample exhibits a significantly larger scatter, particularly at low accretion rates, indicating that DESI probes a wider range of accretion states for a given black hole mass.

3.3. Redshift Dependence of Accretion Properties

Figure 9 presents the redshift dependence of Eddington ratios. SDSS sources predominantly occupy a narrow accretion range across redshift, reflecting its luminosity-selected nature.

DESI, however, reveals a population of low-redshift, low-Eddington-ratio AGNs that are largely absent from SDSS. This difference demonstrates that the apparent redshift evolution of accretion properties is strongly affected by survey selection effects.

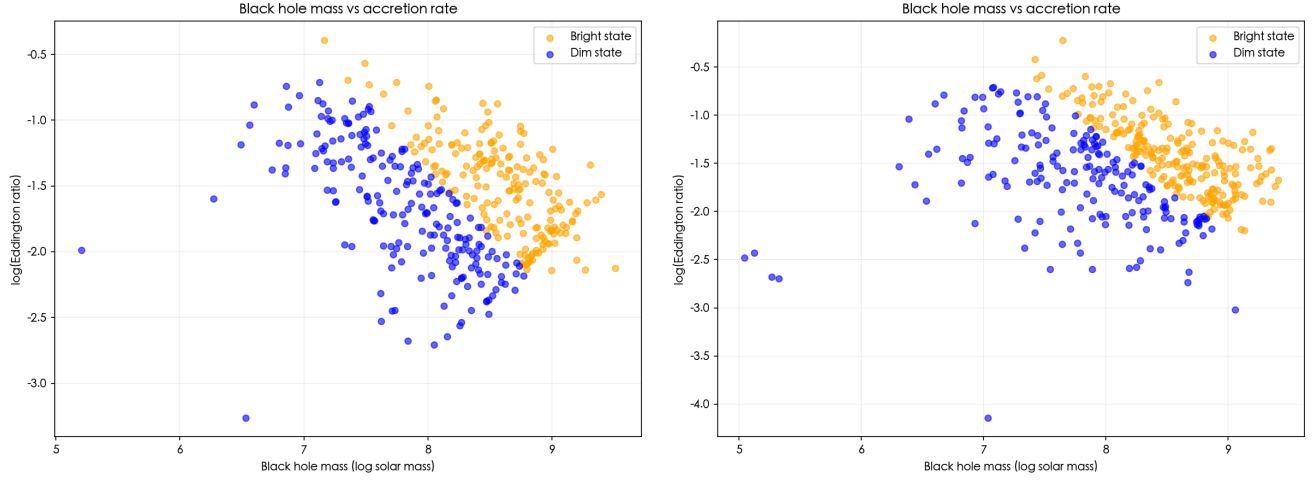


Figure 8. Black hole mass versus Eddington ratio for DESI (left) and SDSS (right) AGNs.

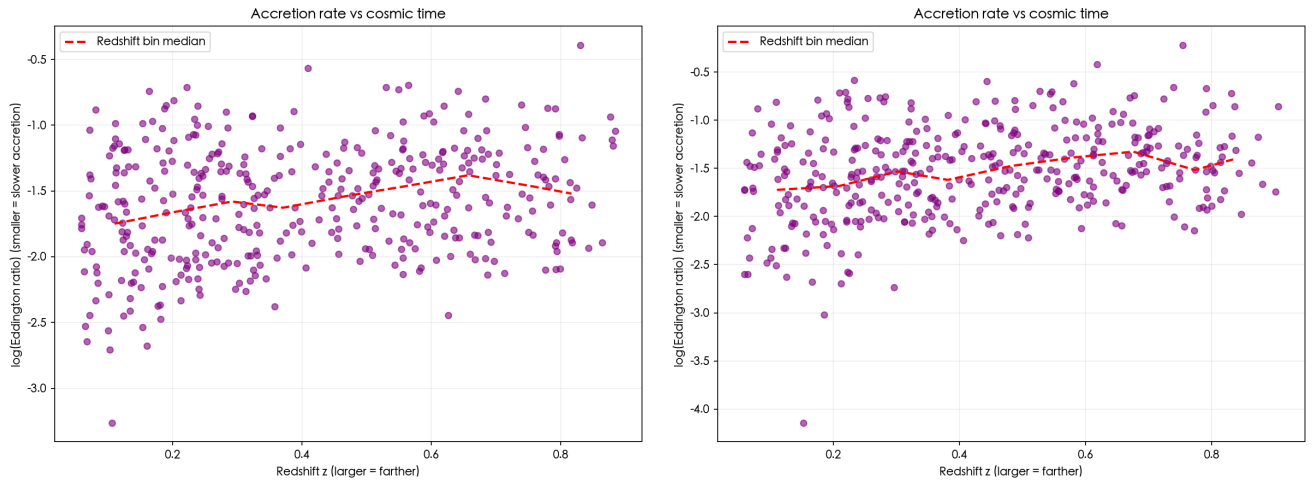


Figure 9. Eddington ratio as a function of redshift for DESI (left) and SDSS (right).

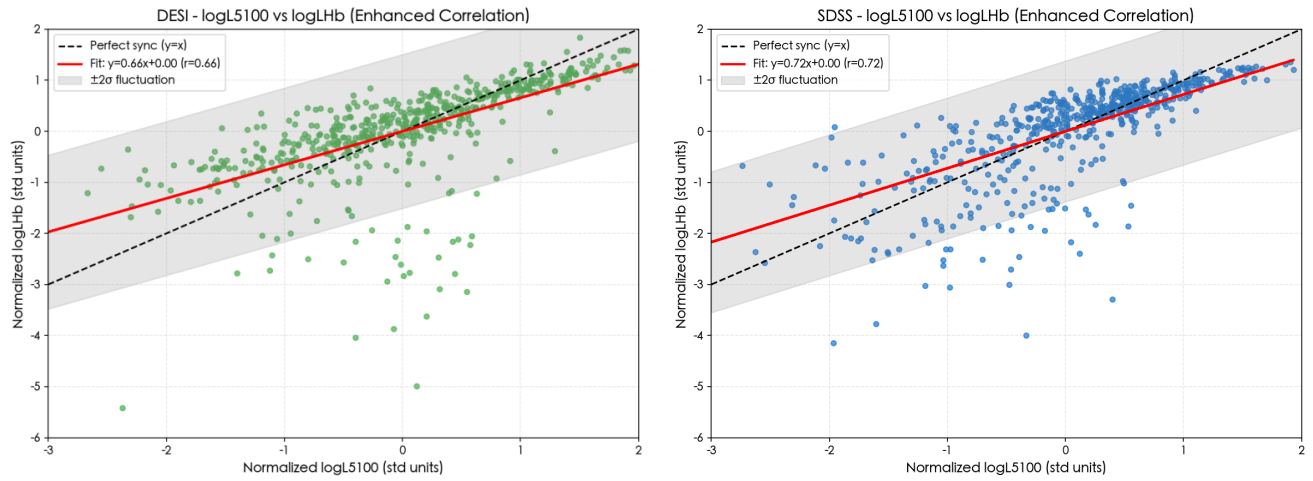


Figure 10. Line luminosity variation proxy versus continuum variation proxy for DESI (left) and SDSS (right).

3.4. Continuum and Line Variability Connection

Figure 10 compares continuum and emission-line variability proxies. Both surveys exhibit a positive correlation, indicating that line variability is driven by changes in the ionizing continuum.

DESI shows a broader dispersion, suggesting that it captures AGNs with weaker or delayed BLR responses, possibly due to larger BLR sizes or lower accretion efficiency.

3.5. Cluster Dependence of Accretion Rate

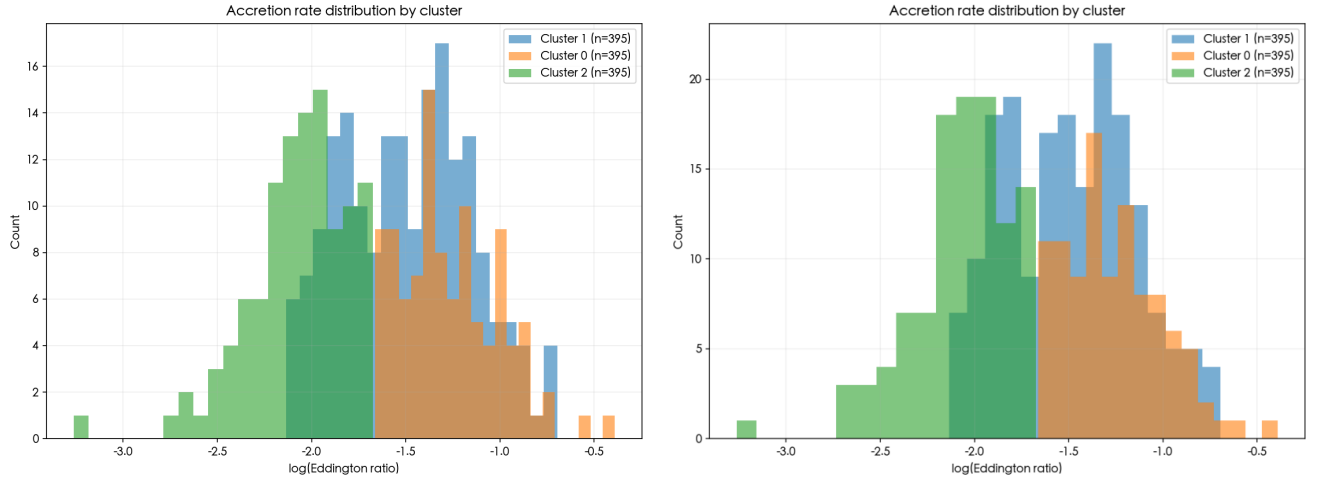


Figure 11. Eddington ratio distributions for different clusters in DESI (left) and SDSS (right).

Figure 11 shows the distribution of Eddington ratios for different clusters. In both surveys, Cluster 2 corresponds to the lowest accretion-rate population.

However, DESI Cluster 2 spans a much wider range in Eddington ratio, extending from $\log \lambda_{\text{Edd}} \sim -4$ to near unity. This indicates that DESI clusters group objects by spectral similarity rather than strict accretion homogeneity.

3.6. Cluster Dependence of Continuum Luminosity

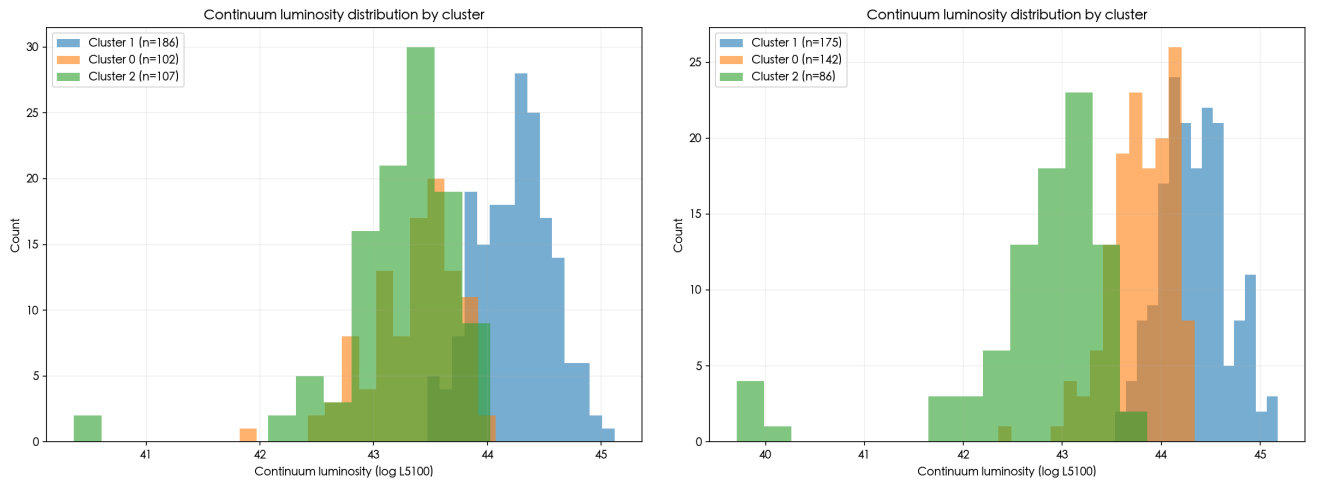


Figure 12. Continuum luminosity distributions for different clusters in DESI (left) and SDSS (right).

Figure 12 further demonstrates that Cluster 2 objects preferentially occupy the low-luminosity regime. DESI again reveals a significantly broader distribution, reinforcing the conclusion that it probes a more diverse and evolutionarily complex AGN population.

4. DISCUSSION

In this section, we synthesize the observational results presented above and discuss their physical implications. By jointly analyzing all ten paired diagnostic figures, we aim to establish a coherent picture of AGN accretion states and their observational manifestations in the DESI and SDSS surveys.

4.1. Universality of Photoionization in the Broad-Line Region

The tight correlations observed between continuum luminosity (L_{5100}) and Balmer-line luminosities ($H\alpha$ and $H\beta$; Figures 1 and 2) provide strong evidence that the broad-line region (BLR) is primarily powered by photoionization from the central accretion disk.

Across both DESI and SDSS samples, the near-linear scaling relations indicate that, despite differences in survey depth and target selection, the fundamental BLR structure remains remarkably stable. This universality suggests that the ionizing spectral energy distribution and BLR gas properties are largely insensitive to survey-specific observational biases. The photoionization equilibrium governs the physical structure of the BLR. Under the reverberation mapping scaling relation, the characteristic radius of the BLR, R_{BLR} , scales with the continuum luminosity as:

$$R_{\text{BLR}} \propto L_{5100}^{0.5}$$

As DESI probes lower luminosities (L_{5100}), the expected R_{BLR} decreases (M. Elitzur 2014; J. R. Trump et al. 2011). The increased scatter observed in our results suggests that near the low-luminosity end, this simple scaling relation may break down as the BLR approaches the condensation radius or becomes dynamically unstable (J. C. Runnoe et al. 2016; Q. Yang et al. 2018). The slightly increased scatter observed in DESI, particularly at low luminosities, can be attributed to host-galaxy contamination, reduced signal-to-noise ratios, and intrinsic variability. These effects become increasingly important when probing weakly accreting or transitioning AGNs.

4.2. Accretion Rate Distributions and Selection Effects

The Eddington ratio distributions (Figure 6) reveal one of the most significant contrasts between DESI and SDSS. SDSS AGNs cluster around moderate accretion rates, consistent with traditional quasar selection techniques that favor luminous, actively accreting systems.

In contrast, DESI spans an exceptionally wide dynamic range, extending down to $\log(L_{\text{bol}}/L_{\text{Edd}}) \sim -4$. This low-accretion tail is of particular physical interest. The distribution peaks significantly below the typical range of SDSS quasars, populating the regime $\log \lambda_{\text{Edd}} < -2$. Theoretical models predict that the transition from a geometrically thin, optically thick disk to a radiatively inefficient accretion flow (RIAF) occurs below a critical Eddington ratio. Following the disk-wind scenario (e.g., Nicastro 2000), the BLR cannot be sustained when the accretion rate falls below:

$$\dot{m} < \dot{m}_{\text{crit}} \approx C \left(\frac{M_{\text{BH}}}{10^8 M_{\odot}} \right)^{-1/8}$$

where the constant C is typically $\sim 10^{-2} - 10^{-3}$. The extended tail of the DESI distribution deeply penetrates this critical regime (J. R. Trump et al. 2011; F. Nicastro (2000), providing a statistical laboratory for testing these instability criteria. Consequently, the DESI sample does not merely reflect a 'fainter' population, but likely contains a high fraction of AGNs undergoing the transition from a bright 'on' state to a dim 'off' state.

4.3. Physical Interpretation of Accretion-State Clusters

The cluster-based analysis of Eddington ratios (Figure 7) provides additional insight into AGN accretion states. Cluster 2, characterized by the lowest average Eddington ratios, shows markedly different behaviors in DESI and SDSS.

In SDSS, Cluster 2 occupies a relatively narrow parameter space, suggesting a more homogeneous population of low-accretion AGNs. By contrast, DESI Cluster 2 extends over nearly three orders of magnitude in Eddington ratio. The "sandwich" accretion disk model can explain the disappearance of the broad-line region and light variation characteristics of DESI's low accretion rate AGN (Q. Ma et al. 2025).

This wide dispersion implies that DESI Cluster 2 is not a uniform population dominated by measurement noise, but a physically heterogeneous group. The properties of Cluster 2 strongly resemble the ‘Fading AGN’ subclass defined by (W.-J. Guo et al. 2025). These objects are characterized by a monotonically dimming light curve over decades and the eventual disappearance of broad emission lines, indicative of a real transition in the accretion disk structure rather than stochastic variability (S. Gezari et al. 2021; M. J. Graham et al. 2023b). Unlike transient flares or TDEs, these sources represent a long-term shutdown of the central engine, consistent with the low Eddington ratios ($\lambda_{\text{Edd}} < 1\%$) observed in our cluster analysis (H. Noda & C. Done 2018; N. P. Ross et al. 2018).

4.4. Implications for AGN Evolution

Taken together, the ten paired diagnostic figures support a picture in which AGNs undergo a continuous range of accretion states rather than occupying discrete classes. DESI’s expanded parameter space reveals a substantial population of low-activity AGNs that were previously inaccessible.

These sources likely represent critical phases in AGN evolution, including shutdown and re-ignition episodes, which are essential for understanding black hole–galaxy co-evolution. Future time-domain and multi-wavelength studies will be crucial for disentangling intrinsic accretion variability from observational biases.

5. CONCLUSIONS

In this work, we performed a comprehensive comparative study of active galactic nuclei observed by DESI and SDSS, making use of ten pairs of diagnostic figures that jointly probe continuum emission, broad-line properties, black hole mass, and accretion efficiency. By systematically analyzing these observables, we aimed to assess both the universality of AGN physical mechanisms and the impact of survey selection effects.

We find that strong, near-linear correlations between the optical continuum luminosity at 5100 Å and the luminosities of broad Balmer emission lines are present in both DESI and SDSS samples. These relations confirm that photoionization by the central accretion disk is the dominant mechanism powering the broad-line region, and that the basic BLR structure is largely independent of survey strategy. However, DESI consistently exhibits larger scatter in these relations, especially at low luminosities, reflecting its sensitivity to fainter AGNs and to systems affected by host-galaxy contamination and lower signal-to-noise ratios.

A key result of this study is the markedly different Eddington ratio distributions revealed by the two surveys. While SDSS AGNs predominantly occupy a relatively narrow range of moderate accretion rates, DESI uncovers a substantial population of weakly accreting sources, extending down to $\log(L_{\text{bol}}/L_{\text{Edd}}) \sim -4$. This low-accretion tail, together with the broad dispersion observed in cluster-based accretion diagnostics, indicates that DESI captures a heterogeneous population of AGNs that were largely absent in previous spectroscopic surveys.

The cluster analysis further suggests that objects with similar spectral characteristics do not necessarily share a uniform accretion state. In particular, the lowest-accretion cluster in DESI spans nearly three orders of magnitude in Eddington ratio, implying that this group likely includes fading AGNs, systems transitioning toward quiescence, and objects whose accretion properties are subject to enhanced measurement uncertainties. Such diversity highlights the importance of deep, wide-area surveys for tracing the full evolutionary sequence of black hole growth. Specifically, the identification of the low-Eddington Cluster 2 provides observational constraints on the population density of AGNs near the BLR disappearance threshold.

Overall, our results demonstrate that DESI significantly extends the observable parameter space of AGN accretion beyond that probed by SDSS. By revealing a large population of low-luminosity and weakly accreting AGNs, DESI provides critical constraints on the late stages of AGN evolution and the duty cycle of black hole activity. Future multi-wavelength and time-domain observations will be essential for disentangling intrinsic accretion variability from observational effects and for establishing a more complete picture of black hole–galaxy co-evolution (D. M. Alexander & R. C. Hickox 2012; R. C. Hickox & D. M. Alexander 2018).

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